

VisionSlot

A Project Report

Submitted in partial fulfilment of the
Requirements for the award of the Degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

By

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Under the esteemed guidance of

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**DEPARTMENT OF INFORMATION TECHNOLOGY ST. ROCK'S
COLLEGE OF ARTS, COMMERCE AND SCIENCE**

(Affiliated to University of Mumbai)

MUMBAI- 400092 MAHARASHTRA

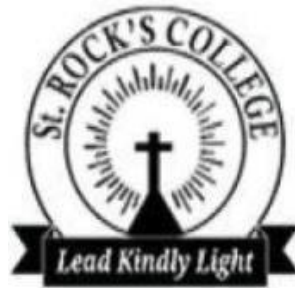
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CERTIFICATE

This is to certify that the project entitled, “VisionSlot” is bonafide work of Alister Darryl Quinny bearing Seat No: 3010913 submitted in partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE in INFORMATION TECHNOLOGY from University of Mumbai.

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ABSTRACT

VisionSlot is an IoT-based smart parking system developed to solve the common problem of parking management in urban areas. The project aims to make parking easier, faster, and more efficient by automatically detecting available parking spaces and controlling entry barriers. Traditional parking systems often require manual monitoring, which can waste time and cause congestion. VisionSlot uses IR proximity sensors to detect vehicles, a servo motor to operate the barrier, and a 20x4 LCD display to show real-time information about available parking spots. All these components are controlled using an Arduino UNO, which acts as the main controller of the system. Key achievements of the project include the development of a simple yet effective system for automated parking management, integration of sensors and display for real-time monitoring, and creation of a user-friendly interface. VisionSlot is useful for parking lots, commercial areas, and residential complexes that want to save time and make parking easier. Future improvements could include mobile app support, larger parking area monitoring, and better connectivity for even smarter management.

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to everyone who contributed to the successful completion and implementation of the VisionSlot project. The journey of building this prototype, from initial circuit design to final VisionSlot testing, has been an enriching experience made possible by the unwavering support of several individuals.

First and foremost, I am profoundly thankful to Ms. Palak Agrawat (HOD) for her invaluable leadership and for providing the academic platform necessary to execute this project. Her vision and management of the department have been a constant source of inspiration throughout this academic endeavour

I extend my sincere thanks to my project guide, Mr. Ameya Rane (Lecturer), for his technical expertise, constant mentorship, and encouragement. His deep insights into Arduino architecture and autonomous systems were crucial in overcoming complex challenges, particularly during the integration of PWM speed optimization.

Finally, I am incredibly thankful to my family and friends for their unwavering support and patience, which served as a constant source of motivation. Their encouragement was vital during intensive hardware debugging phases, such as resolving sensor synchronization issues and perfecting the unique that defines this project.

DECLARATION

I hereby declare that the project entitled, “VisionSlot Parking System” done at Mumbai, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university. The project is done in partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY) to be submitted as final semester project as part of our curriculum

Alister Darryl Quinny

Name and Signature of the Student

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Introduction

1.1 Background

In recent years, the rapid growth of urbanization and the increasing number of vehicles have created significant challenges in parking management. Finding available parking spaces in crowded cities is often time-consuming, leads to traffic congestion, and causes unnecessary frustration for drivers. Traditional parking systems, which rely on manual monitoring and human intervention, are not only inefficient but also prone to errors, resulting in underutilized parking areas and increased operational difficulties. Despite the availability of technology, many parking facilities continue to depend on outdated methods for monitoring occupancy and controlling entry, which fail to provide real-time information to drivers. This has created a growing need for smart solutions that can automate parking management, improve efficiency, and provide accurate, instant updates about space availability. To address this need, VisionSlot was developed as an IoT-based smart parking system that integrates vehicle detection, barrier control, and real-time information display into a single platform. By using IR proximity sensors to detect vehicles, a servo motor to control entry barriers, and a 20x4 LCD display to show available parking spots, VisionSlot aims to reduce manual work, save time, and provide a convenient and user-friendly solution for urban parking facilities.

1.2 Aim and Objectives

Aim

The primary aim of this project is to design and implement a smart, automated parking management system—VisionSlot—built on the Arduino platform. The system will be capable of detecting available parking spots using IR sensors, controlling entry barriers with a servo motor, and displaying real-time parking availability on an LCD screen. The ultimate goal is to provide a cost-effective, reliable, and user-friendly solution to improve parking efficiency and safety in urban areas.

Objectives

The following objectives outline the steps necessary to achieve the project's aim. The VisionSlot system focuses on core system integration by interfacing the Arduino microcontroller with essential hardware components such as IR sensors, servo motors, and an LCD display. This integration ensures seamless data processing, allowing sensor inputs to trigger accurate responses from the barrier mechanism and display system. The project also emphasizes the development of a real-time parking detection system using IR sensors, which reliably identifies vehicle presence in parking slots. Microcontroller algorithms are designed to ensure immediate response when a spot is occupied or vacated, updating the LCD display and controlling the barrier accordingly. Furthermore, automated control is implemented by programming the servo motor to function as a barrier that opens when a vehicle is detected and closes when the spot becomes vacant. A cohesive control system is developed to ensure that the barrier and display operate synchronously, minimizing errors and delays. In addition, system performance and validation are carried out by testing accuracy under various real-world conditions, including different vehicle sizes and lighting environments. Optimization is performed in terms of component selection and coding to achieve low power consumption, durability, and long-term reliability. Overall, the VisionSlot system is evaluated for its efficiency and effectiveness in managing parking spaces while enhancing user convenience.

1.3 Purpose, scope and Applicability

1.3.1 Purpose

The main purpose of this project extends beyond merely assembling a hardware kit; it is to design, implement, and validate a fully functional prototype of an automated smart parking system. The VisionSlot system aims to achieve several critical outcomes in modern parking management and urban IoT solutions. It enables efficient real-time parking detection and management by automatically identifying vacant and occupied parking spots using IR proximity sensors. By continuously monitoring vehicle presence, the system updates the number of available spaces on the LCD display instantly, helping drivers locate parking quickly and reducing congestion within parking areas. Additionally, the system incorporates automated barrier control using a servo motor, ensuring that vehicles can only enter when a parking slot is available. This enhances safety, prevents unauthorized parking, reduces accidents, and maintains an organized flow within the parking area. The project also focuses on improving

user experience and convenience by providing a simple and intuitive interface through the LCD display, allowing users to easily understand parking availability at a glance. Furthermore, VisionSlot holds significant educational value as it demonstrates the integration of microcontrollers, sensors, actuators, and automation, serving as a practical example for students and hobbyists. Lastly, the project promotes smart city and IoT solutions by showcasing how low-cost systems can address urban challenges such as parking inefficiencies, traffic congestion, and space optimization, while offering a scalable approach for future development in smart infrastructure.

1.3.2 Scope

The scope of this project is carefully defined to ensure that all objectives are achievable within the constraints of time, budget, and the capabilities of the Arduino platform. The VisionSlot system is developed using the Arduino UNO microcontroller, which limits processing power, memory, and the number of sensors that can be interfaced. Vehicle detection is carried out using IR proximity sensors placed at designated parking spots and the entry point. The barrier mechanism is controlled by a single servo motor, which operates the gate by raising or lowering it when a vehicle is detected, without incorporating advanced safety features such as weight sensors or anti-collision systems. The mechanical design is limited to a small-scale or indoor parking prototype and is not intended for large-scale, multi-level, or outdoor parking environments with varying terrain.

In terms of system and operational constraints, the system is designed to perform real-time detection and display of available parking slots only. Features such as automated ticketing, payment systems, or cloud-based monitoring are beyond the current scope. The IR sensors provide basic detection of vehicle presence and do not distinguish between different vehicle sizes or types. The entire system operates on a wired setup connecting sensors, servo motor, and LCD display, with no integration of mobile applications, cloud services, or IoT connectivity at this stage. The barrier control logic is purely reactive, meaning the gate operates only in response to sensor input without the use of predictive or adaptive algorithms.

From an environmental and usage perspective, VisionSlot is intended for controlled environments such as indoor parking lots, garages, or small commercial spaces. Its performance may be affected under extreme lighting conditions, outdoor environments, or harsh weather, which can impact the accuracy of IR-based detection. The system is primarily

designed as a small-scale prototype for educational and demonstration purposes, although it has the potential to be scaled and further developed into a real-world smart parking solution.

1.3.3 Applicability

The VisionSlot system is highly applicable across various domains due to its simple yet effective approach to automated parking management. It has strong educational applicability, serving as an excellent hands-on project for students studying IoT, embedded systems, and automation, as it demonstrates sensor integration, microcontroller programming, and actuator control in a real-world scenario. The system is also ideal for small parking areas such as apartments, offices, and commercial complexes where parking spaces are limited, helping drivers quickly locate available spots, thereby saving time and reducing frustration. In the context of smart cities and urban applications, VisionSlot acts as a low-cost IoT solution that can help reduce congestion, improve traffic flow, and optimize parking space utilization.

Furthermore, the project provides a foundation for future scalability, enabling the integration of advanced features such as cloud connectivity, mobile application monitoring, automated payment systems, and data analytics for better urban planning. Although developed as a prototype, VisionSlot serves as a proof of concept for industrial and commercial use, as it can be adapted for applications such as automated garage management, commercial parking systems, and warehouse vehicle entry control. The system also promotes safety and efficiency by automating barrier control and monitoring parking spaces, thereby minimizing human error, reducing the risk of accidents, and ensuring smooth vehicle movement within the parking area. Overall, its versatility, low cost, and ease of implementation make VisionSlot an effective demonstration of how IoT technologies can be applied to solve practical, real-world problems while also serving as a valuable learning tool for future engineers and developers.

Survey of Technologies

2.1 Existing System

Currently, various parking management systems exist, but they often suffer from significant limitations that reduce their efficiency, affordability, and adaptability. Existing parking solutions generally fall into two categories: fully manual systems that rely on attendants or ticket-based operations, and high-end automated systems that are expensive and complex to implement. Many of these systems lack proper automation and real-time feedback, as they either depend on human monitoring or provide automated detection without user-friendly display, making it difficult for drivers to instantly identify available parking spaces and increasing the time spent searching. Additionally, there is a lack of cost-effective IoT-based solutions, since most advanced parking systems are designed for large-scale commercial use and require expensive sensors, cameras, and software, making them unsuitable for small parking areas, apartments, or educational purposes.

Furthermore, existing systems often have low customizability and scalability, as they are rigid in design and do not allow easy integration with IoT technologies or expansion for additional features such as mobile notifications or multi-slot monitoring. These limitations highlight the clear need for a low-cost, real-time, and flexible IoT-based parking solution like VisionSlot. The proposed system effectively bridges the gap between manual and complex automated systems by offering an affordable, adaptable, and practical approach to smart parking, while also serving as a valuable educational tool.

2.2 Proposed System (VisionSlot)

This project aims to overcome the limitations of existing parking solutions by providing a low-cost, automated, and user-friendly system. The VisionSlot system offers real-time parking detection and automated control by integrating IR sensors for vehicle detection with a servo motor for barrier operation. This ensures that only available parking spots are accessible, reducing congestion and improving safety, while simultaneously providing instant updates of vacant spaces through the LCD display. In addition, the system enhances usability and accessibility by offering a simple and intuitive interface, allowing drivers to easily understand parking availability without requiring any technical knowledge or additional training.

Furthermore, VisionSlot is cost-effective, customizable, and educational in nature, as it is built

entirely on the Arduino platform, making it affordable for students, hobbyists, and small-scale applications. Its modular design allows for easy expansion and modification, such as adding more sensors, integrating mobile notifications, or scaling the system for larger parking areas. Overall, the system successfully bridges the gap between affordability and functionality, making it a practical and efficient solution for modern parking management while also serving as a valuable learning tool.

Requirements and Analysis

3.1 Problem Definition

In urban areas, the rapid increase in the number of vehicles has led to a significant shortage of parking spaces, causing drivers to spend excessive time searching for available spots. This not only results in traffic congestion and fuel wastage but also contributes to increased air pollution and driver frustration. The lack of an efficient parking management system makes it difficult to monitor space availability and control vehicle entry in real-time.

The VisionSlot system aims to solve this issue by providing an automated and intelligent parking management solution using IoT technology. The project focuses on developing a cost-effective and user-friendly system that detects the availability of parking slots using IR sensors, displays real-time information on an LCD screen, and automates barrier control through a servo motor. The integration of these components ensures smooth vehicle flow, minimizes human intervention, and optimizes space utilization.

The key challenge lies in synchronizing sensor inputs with mechanical components and display outputs in real time while maintaining system reliability and efficiency. By addressing these issues, VisionSlot provides an innovative step toward creating smarter and more sustainable urban parking systems.

3.2 Requirement Specification

3.2.1 Functional Requirements

The functional requirements define the essential operations that the VisionSlot system must perform to achieve its objectives. These requirements describe the specific features and interactions between the hardware and software components. The system must be capable of detecting the presence or absence of a vehicle using IR sensors placed at each parking slot. Based on this detection, the LCD display should show the current status of all parking slots, such as P1: F, P2: E, P3: F, and update automatically whenever a slot becomes occupied or vacant.

In addition, the system must provide automatic barrier control, where the entry IR sensor detects an arriving vehicle and triggers the servo motor to open the entry barrier. After the vehicle passes, the exit IR sensor ensures the barrier is closed. The Arduino microcontroller

should continuously monitor all IR sensors to enable real-time updates of slot availability and system status. During system initialization, all sensors must be properly set up, the servo motor should move to its default position (barrier closed), and the LCD display should be cleared for fresh output. Furthermore, the system should include basic error handling to manage sensor failures or incorrect readings by maintaining a stable display and preventing unintended servo movements.

3.2.2 Non-Functional Requirements

The non-functional requirements define the quality attributes and constraints that ensure the VisionSlot system operates efficiently and reliably. The system must maintain high reliability by providing accurate detection and display with minimal false readings under normal operating conditions. It should also deliver good performance, ensuring that all sensors and the servo motor respond within a short delay of less than one second to enable smooth vehicle entry and exit. Scalability is an important factor, as the system design should allow the addition of more parking slots with minimal changes in code and wiring.

In terms of usability, the LCD interface should present slot statuses in a clear and readable format that can be easily understood by non-technical users. The system must also be power efficient, operating effectively on a 5V supply from the Arduino Uno without causing overheating or overloading of components. Maintainability is ensured by keeping the wiring and code structure simple, allowing for easy debugging, maintenance, and replacement of sensors when required. Additionally, the system should be portable, as the use of Arduino Uno and IR sensors makes it compact and suitable for small-scale parking setups, demonstrations, and educational purposes.

3.3 Planning and Scheduling

3.3.1 Gantt Chart

A Gantt chart provides a visual timeline for the tasks in the project, showing the planning and scheduling of entire project.

Figure 1 GANTT CHART

Figure 2 GANTT CHART

3.3.2 PERT Chart

A PERT (Program Evaluation and Review Technique) Chart helps visualize the project workflow, dependencies, and estimated completion times. Below is a breakdown of the project in terms of tasks and dependencies.

Figure 3 PERT CHART

3.4 Software and Hardware Requirements

3.4.1 Software Requirements

1. **Microcontroller Programming:** Arduino IDE – To write, compile, and upload code to the Arduino UNO. Supports C/C++ programming for controlling sensors, servo motor, and LCD.
2. **Control and Display Logic:** Embedded C/C++ Libraries – For interfacing with IR sensors, servo motor, and 20x4 LCD. Examples include Servo.h and LiquidCrystal_I2C.h.
3. **Communication:** Bluetooth Communication Libraries – To establish serial communication between Arduino and mobile app (optional, if future expansion includes remote monitoring).
4. **Simulation and Testing Tools:** Proteus or Tinker cad Circuits (Optional) – For simulating the circuit before physical assembly.
5. **Mobile App / User Interface (Optional for Expansion):** MIT App Inventor or Arduino Bluetooth Controller App – For basic monitoring or manual control if the system is upgraded to remote access.
6. **Development Environment:** Windows, Linux, or macOS – For running the Arduino IDE and simulation software.
7. **USB Drivers** – To ensure proper communication between the PC and Arduino UNO.

3.4.2 Hardware Requirements

1. **Microcontroller: Arduino UNO R3** – Serves as the main controller for the system, processing sensor data and controlling the barrier and display.
2. **Barrier Control: SG90 Servo Motor** – Operates the entry barrier to raise and lower based on parking spot availability.
3. **Vehicle Detection Sensors: IR Proximity Sensors** – Detect vehicles at parking spots and at the entry point.
4. **Display System: 20x4 LCD with I2C Module** – Shows the number of available parking spots in real-time.
5. **Power Supply: 9V Batteries (x2) or equivalent DC power supply** – Powers the Arduino, sensors, and servo motor.
6. **Battery Connectors and Power Switch** – For connecting and controlling the power to the system.
7. **Additional Components: Jumper wires, resistors, and breadboard** – For making electrical connections between components.

3.5 Conceptual Model

3.5.1 Block Diagram

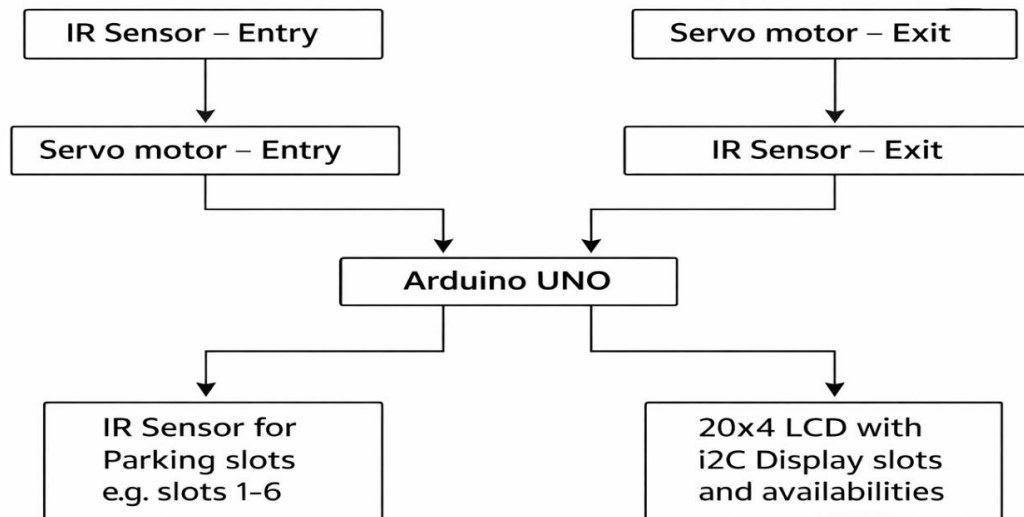


Figure 4 Block Diagram

3.5.2 Class Diagram

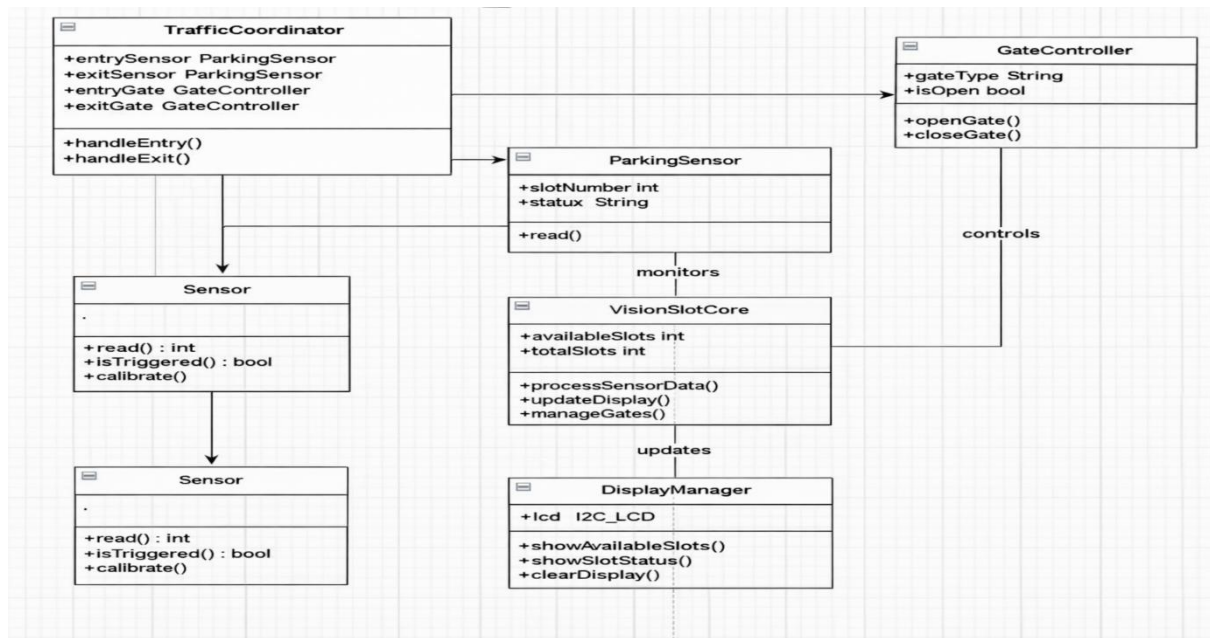


Figure 5 Class Diagram

3.5.3 Flowchart

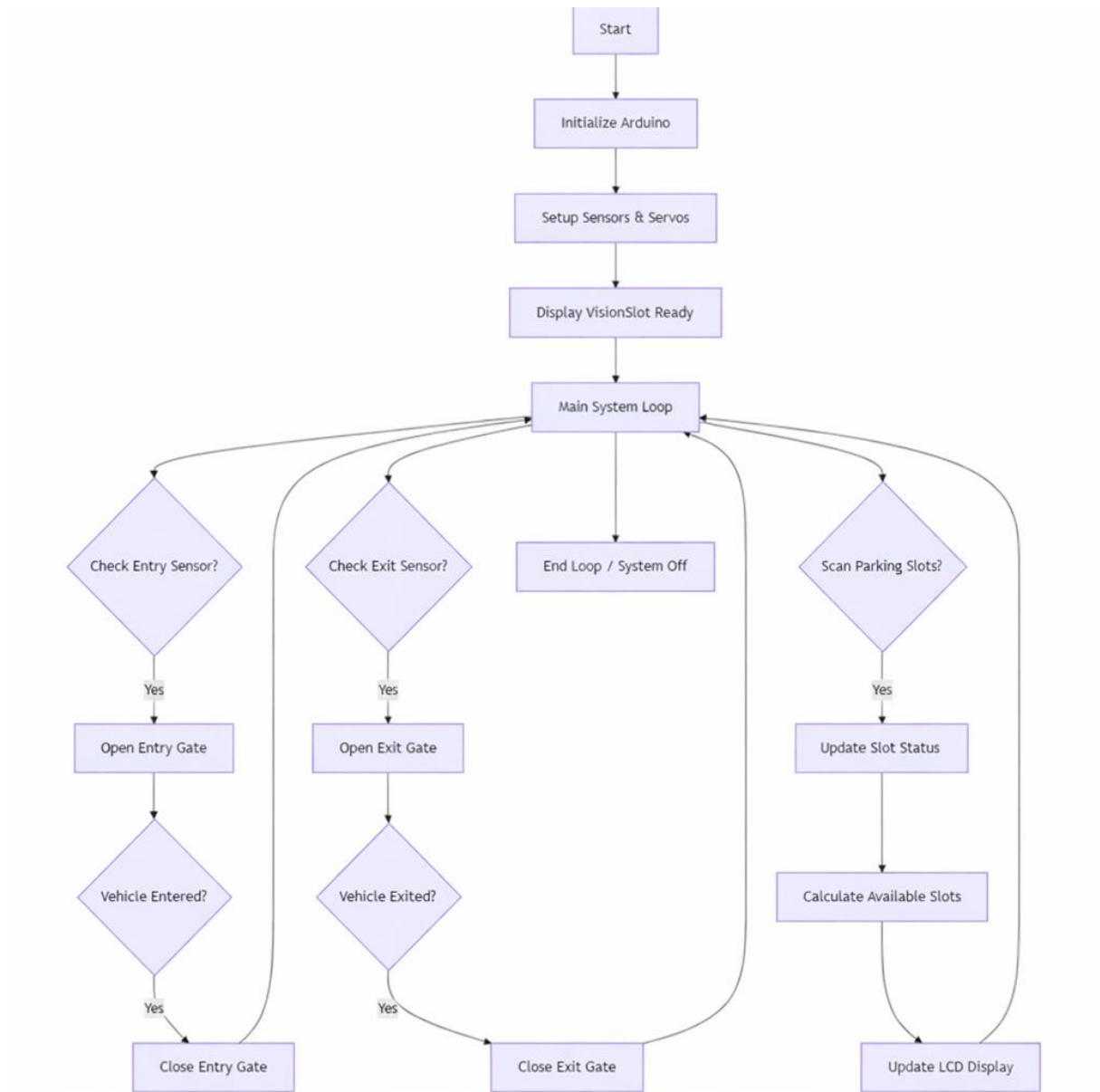


Figure 6 Flowchart

3.5.4 Circuit Diagram

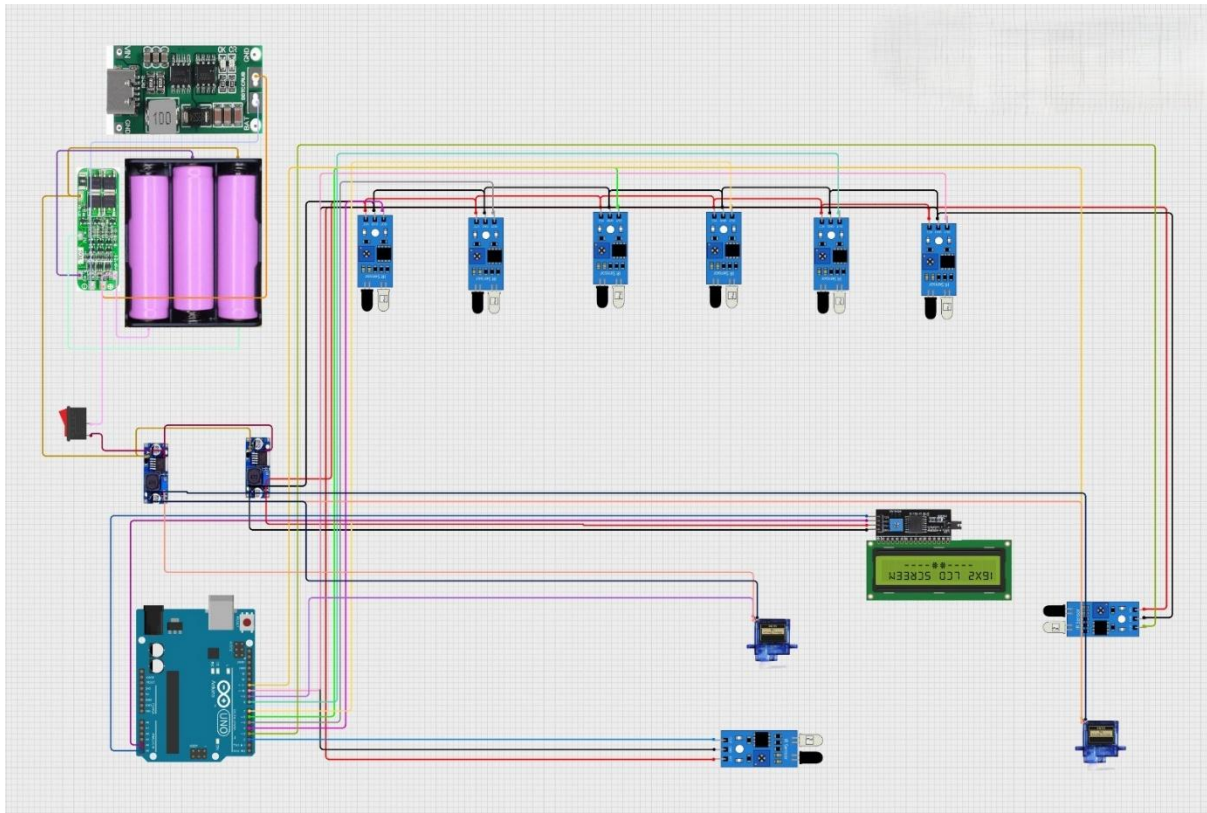


Figure 7 Circuit Diagram

Pins Allocations:

Ir sensors (Parking Slots)

Slot	Arduino pin
P1	D4
P2	D5
P3	D6
P4	D7
P5	D8
P6	D10

IR Sensor (Entry and Exit):

Function	Arduino Pin
Entry Ir sensor	D2
Exit Ir sensor	D3

Servo Motor :

Function	Arduino Pin
Entry Servo	D9
Exit Servo	D11

Lcd Display (i2c):

LCD Pin	Arduino Pin
SDA	A4
SCL	A5

Power Connections:

Component	Connections
Arduino	5V/Vin
Sensors	5V & GND
Servos	5V & GND
LCD	5V & GND

System Design

4.1 Basic Modules

Entry Detection Module

This module detects the presence of a vehicle at the entry gate. It consists of an IR proximity sensor placed before the servo-controlled barrier.

Function: Detects an approaching vehicle.

Operation: When a car is detected by the IR sensor, the Arduino processes the signal and sends a command to open the gate via the servo motor.

Objective: To automate gate opening and reduce manual intervention.

Exit Detection Module

This module handles the detection of vehicles leaving the parking area. It uses another IR sensor placed after the barrier.

Function: Detects vehicle departure.

Operation: Once the car exits, the Arduino triggers the servo to close the barrier.

Objective: To ensure the gate closes automatically after a vehicle passes through and updates the available slot count.

Parking Slot Monitoring Module

This is the core module of the system responsible for monitoring the occupancy status of each parking slot using IR sensors.

Function: Detects whether each slot is occupied or vacant.

Operation: Each IR sensor sends a digital signal to the Arduino, which then processes the data and updates the slot status.

Objective: To maintain an accurate count of available parking spaces.

Display Module

This module uses a 20x4 I2C LCD display to show real-time information about parking availability.

Function: Displays the number of free and occupied slots.

Operation: The Arduino updates the display whenever a vehicle enters or exits the parking area.

Objective: To provide drivers with clear and quick information about available parking spaces.

Control Module

This is the central logic unit of the VisionSlot system, handled by the Arduino Uno microcontroller.

Function: Processes inputs from all sensors and controls outputs such as the LCD display and servo motor.

Operation: Based on sensor data, it decides when to open/close the gate and update slot status.

Objective: To coordinate the entire system for smooth and synchronized operation.

Power Supply Module

This module provides regulated power to all components in the system.

Function: Supplies 5V DC power to the Arduino, sensors, servo motor, and LCD display.

Operation: A regulated adapter or USB power supply is used to ensure stable operation of all components.

Objective: To deliver a steady and safe power source to prevent malfunctioning.

4.2 Procedural Design

The VisionSlot system follows a structured procedural design to ensure smooth operation, efficient data handling, and accurate parking slot monitoring. The design defines the logical flow of data between the sensors, the Arduino controller, and the output display. Each procedure in the system is responsible for a specific function that contributes to the overall working of the project.

The major procedures in the VisionSlot system are as follows:

Initialization Procedure:

This procedure initializes the Arduino, LCD display, and IR sensors.

It sets up input and output pin and displays a welcome message on the LCD when the system starts.

Sensor Reading Procedure:

The IR sensors continuously detect the presence or absence of vehicles in each parking slot. Each sensor outputs a digital HIGH or LOW signal depending on whether a vehicle is detected.

These readings are then sent to the Arduino for processing.

Data Processing Procedure:

The Arduino analyses the sensor inputs to determine whether each parking slot is “Available” or “Occupied.”

The logic compares the signals from all sensors and updates the status of each slot accordingly.

Display Update Procedure:

This procedure updates the 20x4 LCD display with the current parking slot information. The LCD shows messages such as “Parking 1: Occupied” or “Parking 2: Available,” as well as the total number of free slots.

System Monitoring Procedure:

The entire process runs continuously in a loop, ensuring real-time updates of parking availability.

The system can be extended to include additional modules like barrier control, mobile app integration, or cloud-based data storage.

Overall, the procedural design of VisionSlot emphasizes real-time monitoring, accurate sensor input handling, and efficient communication between hardware components. This structured approach ensures reliability, scalability, and ease of future enhancements.

4.3 Algorithm Design

The VisionSlot system follows a step-by-step algorithm to detect vehicle presence, update parking slot availability, and display the status in real-time. The algorithm ensures accurate data collection from IR sensors and smooth communication with the LCD display.

Algorithm Steps:

Start the System: Power on the Arduino and initialize all connected components (IR sensors, LCD, and servo motor if used).

Initialize Components: Set the pin modes for all IR sensors as input.

Set the pin mode for the LCD display as output.

Display a welcome message such as “WELCOME TO VISIONSLOT PARKING.”

Read Sensor Inputs

Continuously read the digital signals from each IR sensor.

For each slot:

If the IR sensor detects an obstacle (LOW signal), mark the slot as Occupied.

If no obstacle is detected (HIGH signal), mark the slot as Available.

Process Data: Count the number of available and occupied slots.

Store this data temporarily in memory for quick updating.

Update LCD Display: Display the real-time parking information such as:

Free Slots: 1

Parking 1: F (Full)

Parking 2: E (Empty)

Continuously refresh the LCD to reflect any changes detected by the sensors.

Repeat Loop: The system continuously repeats the above steps, ensuring real-time updates.

4.4 Code: Parking System

```
void loop()
{
    int s1 = digitalRead(S1);
    int s2 = digitalRead(S2);
    int s3 = digitalRead(S3);
    int s4 = digitalRead(S4);
    int s5 = digitalRead(S5);
    int s6 = digitalRead(S6);

    int available = 0;

    if(s1) available++;
    if(s2) available++;
    if(s3) available++;
    if(s4) available++;
    if(s5) available++;
    if(s6) available++;

    if(digitalRead(IR_ENTRY) == LOW && available > 0)
    {
        entryServo.write(ENTRY_OPEN_ANGLE);
        entryOpen = true;
    }
    else if(digitalRead(IR_ENTRY) == HIGH)
    {
```

```

    entryServo.write(ENTRY_CLOSE_ANGLE);
    entryOpen = false;
}

if(digitalRead(IR_EXIT) == LOW)
{
    exitServo.write(EXIT_OPEN_ANGLE);
    exitOpen = true;
}
else if(digitalRead(IR_EXIT) == HIGH)
{
    exitServo.write(EXIT_CLOSE_ANGLE);
    exitOpen = false;
}

updateScroll();
updateLCD(available, s1,s2,s3,s4,s5,s6);

delay(100);
}

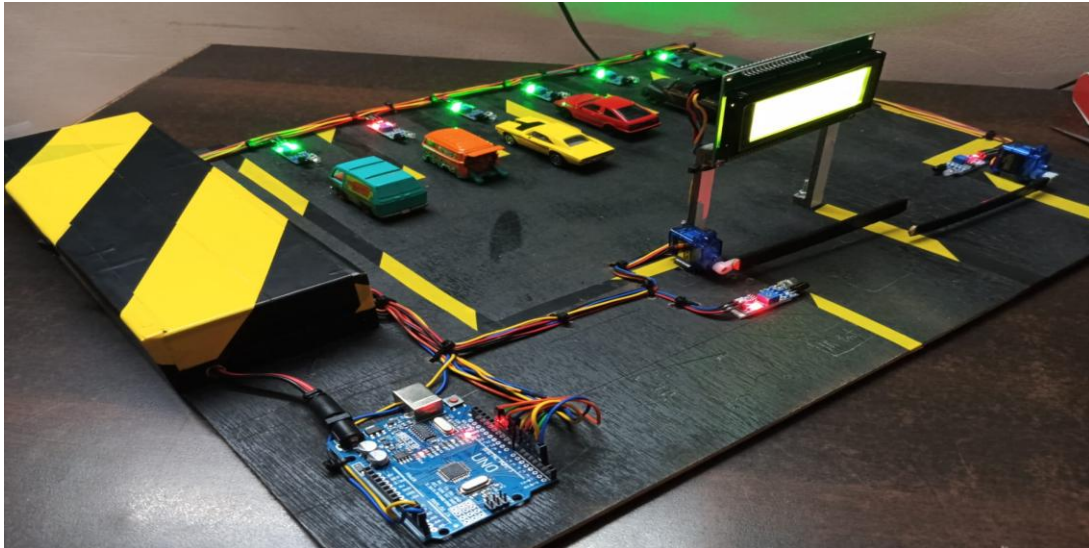
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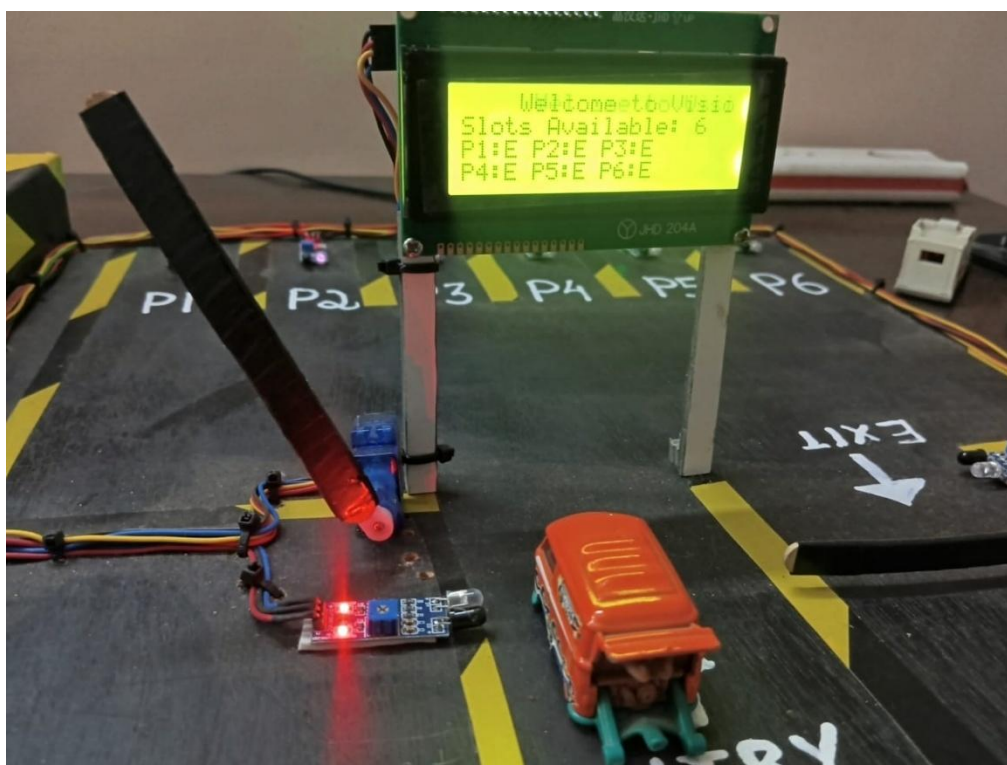
Results and Discussion

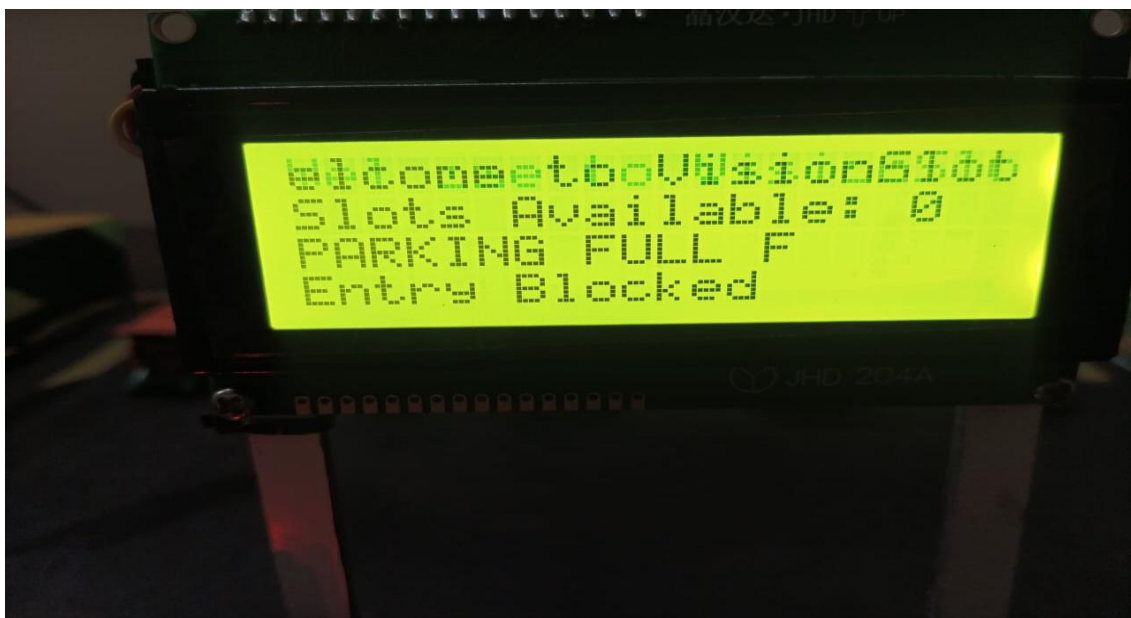
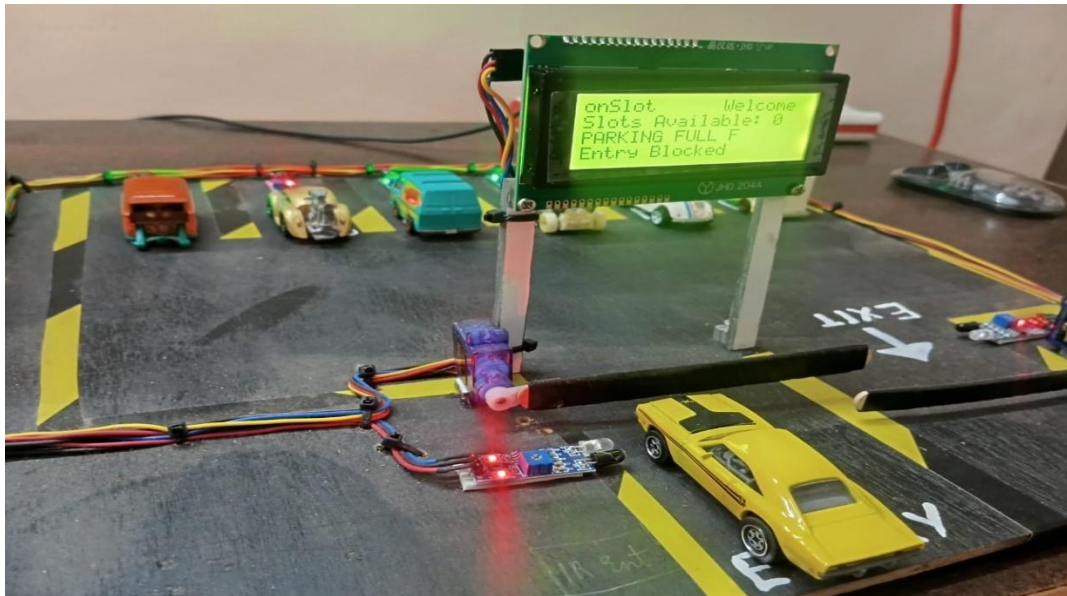
The VisionSlot – IoT Based Smart Parking System was successfully implemented and tested. The system effectively detects vehicle presence using IR sensors and displays real-time parking availability on a 20x4 LCD (I2C).

From the experimental setup and results observed:

1. Initially, when no vehicles are present, the LCD displays “Slots Available: 6”, indicating all parking slots are empty.
2. As a vehicle enters the parking area, the IR sensor detects its presence and updates the slot status on the LCD. For example, when one slot is occupied, it shows “Slots Available: 5” and marks the occupied slot (e.g., P1: F).
3. The system continuously updates all slot statuses (P1 to P6) in real time, showing whether each slot is Empty (E) or Full (F).
4. The servo motor barrier operates correctly:
 - It opens when parking slots are available.
 - It closes (blocks entry) when the parking is full.
5. When all parking slots are occupied, the LCD displays:
 - “Slots Available: 0”
 - “PARKING FULL”
 - “EntryBlocked”indicating that no more vehicles are allowed inside.
6. The system successfully demonstrates:
 - Real-time slot monitoring
 - Automatic barrier control
 - Accurate display of parking status
 - Efficient working of sensors and display module







Advantages and Limitations

6.1 Advantages

The VisionSlot system provides an intelligent and automated solution to modern parking problems by utilizing IoT technology, sensors, and a user-friendly interface. It offers several benefits to users, parking owners, and city management. One of the major advantages of the system is the reduction of traffic congestion, as vehicles no longer need to roam around searching for parking spaces, thereby improving overall traffic flow. The system also provides real-time slot availability by using sensors to instantly update the status of parking spaces, allowing users to quickly identify free and occupied slots.

In addition, VisionSlot contributes to environmental sustainability by reducing unnecessary fuel consumption and lowering air pollution caused by prolonged vehicle movement. The system is designed with a simple and user-friendly interface, making it accessible even to non-technical users without requiring any special training. It also enables efficient space utilization, as parking owners can monitor slot usage patterns and manage parking areas more effectively. Furthermore, the automation of the system significantly reduces human effort, minimizing the need for manual supervision and lowering labour costs. Overall, VisionSlot enhances efficiency, convenience, and sustainability in modern parking management systems.

6.2 Limitations

Although VisionSlot offers numerous advantages, it also has certain limitations and challenges that must be considered. One of the primary concerns is the initial setup cost, as installing sensors, microcontrollers, and other hardware components increases the overall implementation expense, especially when scaling the system to accommodate more parking slots. In addition, sensor-related issues may arise, as IR sensors can sometimes produce inaccurate readings due to dust, environmental conditions, or hardware faults, and improper alignment can further affect detection accuracy. The system also requires regular maintenance to ensure smooth operation, including periodic cleaning, calibration, and timely replacement of components when necessary.

Another important limitation is the system's dependency on a continuous power supply, as any power failure can temporarily disrupt its functioning, and the absence of a backup power source can reduce reliability. Furthermore, VisionSlot is limited to the specific areas where sensors

are installed, making it unsuitable for managing parking in uncovered or large-scale open environments without significant modifications. Expanding the system to larger areas would require additional infrastructure and redesign. Despite these challenges, the system remains effective for small-scale and controlled environments, particularly for educational and demonstration purposes.

Conclusion and Future Work

7.1 Conclusion

The VisionSlot – IoT Based Smart Parking System project was developed to solve the common problem of finding parking spaces in crowded urban areas. In many cities, drivers waste time and fuel searching for parking, which increases traffic congestion and pollution. VisionSlot provides a smart and automated solution using sensors, microcontrollers, and a web-based application.

The system detects vehicle presence using sensors and updates the parking slot status in real time. Users can view available spaces and reserve slots through a simple website interface. This reduces manual effort and makes the parking process faster and more organized.

Overall, VisionSlot proves that IoT technology can be effectively used to build smart city solutions. The project meets its objectives and provides a reliable, user-friendly, and cost-effective parking system.

7.2 Future Work

The VisionSlot system can be further enhanced with several advanced features to improve its functionality, efficiency, and usability in real-world applications. A dedicated mobile application for Android or iOS platforms can be developed to provide users with easier access and a better overall experience. Integration of online payment options such as UPI, credit/debit cards, and digital wallets can enable seamless cashless parking transactions. Additionally, GPS navigation support can be incorporated to guide users directly to the nearest available parking slot, reducing search time and improving convenience. The inclusion of Automatic Number Plate Recognition (ANPR) using cameras can enhance security and enable automated vehicle identification.

Furthermore, cloud database integration can allow large-scale data storage, remote monitoring, and centralized control from anywhere. The system can also be expanded to support multi-level parking structures such as malls and commercial complexes, making it suitable for larger deployments. To promote sustainability, solar power support can be added to reduce dependency on conventional electricity and make the system more eco-friendly. Smart notification features can also be implemented to alert users about booking confirmations,

parking duration, expiry times, and slot availability. These future enhancements will significantly improve the system's intelligence, efficiency, and versatility, making VisionSlot a more robust and scalable solution for diverse real-world applications.

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